

Dialectical Behavior Therapy vs Schema Therapy for Patients With Borderline Personality Disorder

The BOOTS Multicenter Randomized Clinical Trial

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IMPORTANCE Head-to-head comparisons of evidence-based treatments for borderline personality disorder (BPD), including dialectical behavior therapy (DBT) and schema therapy (ST), remain scarce, despite their widely recognized relevance.

OBJECTIVE To compare the effectiveness of DBT and ST in a multicenter trial among outpatients with BPD.

DESIGN, SETTING, AND PARTICIPANTS This 3-year multicenter superiority randomized clinical trial (RCT), part of the BOOTS (Borderline Optimal Treatment Selection) project, was conducted between January 2019 and April 2025. The trial was conducted in 9 Dutch outpatient centers among individuals aged 18 to 65 years with BPD. Assessments were administered by assessors blinded to treatment allocation.

INTERVENTIONS Participants were randomized to DBT or ST via computerized covariate-adaptive randomization accounting for sex and BPD severity. Both treatments were delivered over 2 years in a combined group-individual format.

MAIN OUTCOMES AND MEASURES The primary outcome was the rate of change in *DSM-5* BPD severity from baseline to 1-year posttreatment follow-up (3 years after baseline), assessed by the Borderline Personality Disorder Severity Index, fifth version (BPDSI-5) total score. Secondary outcomes included *DSM-5* BPD criteria, general symptom severity, functioning, quality of life, well-being, sleep problems, and treatment dropout.

RESULTS Of 204 participants (172 female [84.3%]; mean [SD] age, 32.21 [9.57] years), 95 (46.6%) received DBT and 109 (53.4%) received ST. Intention-to-treat analysis revealed no significant difference between DBT and ST in reducing BPD severity ($P = .27$, $r = 0.09$), with large improvements in both treatments. At 36-month follow-up, the estimated mean difference was -1.09 on the BPDSI-5 (Cohen $d = 0.15$; 95% CI, -0.17 to 0.47). No significant differences between treatments were observed on any secondary outcome, with marked improvements over time in both treatments. Dropout did not differ significantly (DBT: 34% at 1 year, 52% at 2 years; ST: 29% at 1 year, 46% at 2 years).

CONCLUSIONS AND RELEVANCE This multicenter RCT found no significant differences in effectiveness between DBT and ST for outpatients with BPD, with substantial improvements across outcomes. Future research is warranted to assess equivalence and to determine which treatment works best for whom and why.

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 Visual Abstract

 Supplemental content

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Borderline personality disorder (BPD) is a significant public health issue,¹ marked by chronic instability in multiple areas.² BPD is associated with high rates of comorbidity, functional impairment, low quality of life, and high suicide risk.^{3,4} Besides the intensive personal suffering, BPD also imposes a significant financial burden on society due to extensive use of psychiatric health care services and psychotropic medication, productivity losses, and reliance on informal care.⁵ While BPD has traditionally been viewed as a difficult-to-treat condition,⁶ several evidence-based treatment approaches have been developed and evaluated, including dialectical behavior therapy (DBT^{7,8}) and schema therapy (ST^{9,10}). DBT is a comprehensive cognitive behavioral treatment rooted in Zen, behaviorism, and dialectical philosophy,¹¹ whereas ST integrates cognitive behavioral and experiential techniques with developmental and psychodynamic concepts.¹² Both treatments demonstrated effectiveness,¹³⁻¹⁵ although the evidence is generally of low to moderate quality,^{16,17} and ST has been studied less extensively than DBT. Moreover, head-to-head comparisons between treatments are scarce, even though the importance of these comparisons has been widely recognized.^{15,17,18} Such comparisons are essential for advancing treatment options for individuals with BPD, enhancing informed decision-making for patients, and guiding health care policymakers in setting priorities for treatment reimbursement.¹⁴

To date, only 1 study has directly compared DBT and ST in a randomized clinical trial (RCT).¹⁴ This study found substantial improvements in multiple domains for both treatments, including BPD symptoms, general symptom severity, psychosocial functioning, and quality of life. In general, no significant differences were found between DBT and ST in treatment outcomes, except for *DSM-5* BPD criterion 8 (anger), in which DBT was superior to ST. Additionally, retention rates did not differ after treatment, although more DBT patients dropped out earlier than ST patients. However, Assmann and colleagues¹⁴ reported on a single-center trial, conducted in a tertiary outpatient center affiliated with inpatient treatment. This center was specifically for patients diagnosed with severe BPD who are frequently seen in emergency departments or would otherwise require inpatient treatment. Both the single-center setting and the specific focus on severely affected patients limit external validity. Accordingly, Assmann and colleagues¹⁴ emphasized the need for additional trials comparing the effectiveness of DBT and ST. The present study is part of the BOOTS (Borderline Optimal Treatment Selection) project¹⁹ and addresses these limitations by comparing the effectiveness of DBT and ST in a multicenter superiority RCT among outpatients with BPD. In addition to BPD severity as the primary outcome, the current trial explored secondary outcomes in different areas, including symptoms, functioning, well-being, and treatment dropout.

Methods

Study Design

This study is a multicenter superiority RCT with 2 active treatment conditions (DBT and ST), conducted between January

Key Points

Question What is the comparative effectiveness of dialectical behavior therapy (DBT) vs schema therapy (ST) for individuals with borderline personality disorder (BPD)?

Findings In this multicenter randomized clinical trial that included 204 adults with BPD, BPD severity decreased markedly over the 3-year study period, with no statistically significant differences in effectiveness between DBT and ST.

Meaning The findings indicate no evidence of superiority of either treatment for outpatients with BPD, with both treatments associated with substantial long-term improvements.

2019 and April 2025. Based on the minimum statistically detectable effect approach,²⁰ the planned sample size of 200 was considered sufficient to detect medium-sized treatment effects, while accounting for the nested structure of the data. This study adheres to the Consolidated Standards of Reporting Trials (CONSORT) guidelines.²¹ A comprehensive description of the study design is available in the published study protocol¹⁹ and the trial protocol (Supplement 1).

Participants

Participants were eligible for inclusion if they (1) were aged 18 to 65 years, (2) had BPD as their primary diagnosis with a Borderline Personality Disorder Severity Index, fifth version (BPDSI-5) score above 20, (3) were proficient in Dutch, and (4) were motivated and available to participate in treatment and assessments. Exclusion criteria included having (1) a substance use disorder requiring clinical detoxification, (2) a psychotic disorder or bipolar I disorder with at least 1 manic episode within the past year, (3) an antisocial personality disorder combined with violent interpersonal behavior in the past 2 years, (4) an IQ below 80, (5) a travel time exceeding 45 minutes to the mental health care center, (6) no permanent address, and (7) received ST or DBT within the past year.

Procedure

The study was registered in the Overview of Medical Research in the Netherlands and approved by the Medical Ethics Committee of the Academic Medical Center Amsterdam. A patient representative contributed to the study design and implementation. Participants were recruited between January 2019 and October 2021 at 9 Dutch mental health care centers: Antes (Rotterdam); GGZ inGeest (Amsterdam); GGZ Noord-Holland-Noord (Heerhugowaard); GGZ Rivierduinen (Leiden); Nederlands Psychoanalytisch Instituut (Amsterdam); Pro Persona (Ede and Tiel); PsyQ (Rotterdam-Kralingen); and PsyQ, i-psy, and the Academisch Centrum voor Trauma en Persoonlijkheid (Amsterdam). After providing written informed consent, individuals with BPD or suspected of having BPD were screened (eMethods in Supplement 2). Eligible participants completed the baseline assessment, including computer-based self-report questionnaires and interviews, and were randomized to DBT or ST by an independent research assistant using computerized covariate-adaptive randomization²²⁻²⁴ to account for sex and BPD severity (BPDSI-5 score <24 vs ≥24).

After the treatment phase started, participants were reassessed every 6 months up to 1 year posttreatment (follow-up). The interviews were conducted by independent assessors who had no role in treatment delivery, trial design, or trial reporting; had no therapeutic relationship with participants; and were blinded to treatment allocation. Further details regarding the procedure can be found in the study protocol.¹⁹

Treatments

Both treatments were manualized, using ST manuals by Farrell and Shaw²⁵ and by Arntz and Van Genderen,⁹ and the DBT manual by Linehan.^{7,8} DBT and ST had a maximum duration of 25 months, including a pretreatment phase (± 4 weeks), a combined group-individual treatment phase (DBT: 12 months, ST: 18 months), and a maintenance phase (DBT: 12 months, ST: 6 months). In DBT, the treatment phase consisted of weekly individual sessions, weekly skills training group sessions, and between-session consultation, followed by a blended after-care program including monthly individual sessions, 3 monthly group sessions, and eHealth (an online platform providing instructional videos, digital worksheets, and therapist feedback).²⁶ In ST, patients received weekly individual and group sessions, with reduced frequency of individual sessions after 12 months, followed by a maintenance phase consisting of biweekly individual sessions for 3 months and subsequently 3 monthly individual sessions. Across both treatments, the total dose was approximately 167 hours. Groups were offered in a semiopen format. All therapists were trained and received supervision. Detailed information is provided in the eMethods in [Supplement 2](#) and the study protocol.¹⁹

Due to the COVID-19 pandemic, protocol deviations (eg, treatment interruptions, online sessions) occurred. The potential influence of these deviations was assessed in sensitivity analyses.

Treatment Integrity

Treatment adherence analyses indicated treatment differentiation for both individual sessions (DBT adherence: $P = .002$, $d = 0.59$; ST adherence: $P < .001$, $d = 1.84$) and group sessions (DBT adherence: $P < .001$, $d = 3.34$; ST adherence: $P < .001$, $d = 3.84$). Interrater agreement (intraclass correlation coefficient [ICC]) was adequate for group sessions (ST adherence: 0.94; DBT adherence: 0.85), but unsatisfactory for individual sessions (ST adherence: 0.34; DBT adherence: -0.06). Additional details are provided in the eMethods in [Supplement 2](#).

Measures

The primary outcome was the rate of change in severity of the DSM-5 BPD manifestations from baseline to follow-up, assessed by the BPDSI-5 total score (range, 0-90).^{27,28} The BPDSI is specifically developed to comprehensively assess BPD symptoms and has shown strong psychometric properties.^{27,28} In our sample, excellent interrater reliability (ICC = 0.98) was observed. Secondary outcomes included the 9 DSM-5 BPD criteria (BPDSI-5^{27,28}), experienced burden of BPD symptoms (Ultrashort BPD Checklist [uBPDC²⁹]), general psychopathological symptoms (Brief Symptom Inventory³⁰), functional

impairment (World Health Organization Disability Assessment Schedule 2.0³¹), generic and disease-specific quality of life (5-level EuroQol 5-dimensional version³² and Mental Health Quality of Life 7-dimensional Questionnaire,³³ respectively), well-being (happiness³⁴ and Outcome Rating Scale³⁵), sleep problems (Insomnia Severity Index³⁶ and Nightmare Frequency Questionnaire³⁷), participation in work or study, treatment dropout, and serious adverse events (SAEs). Details on the measures are provided in the eMethods in [Supplement 2](#).

Statistical Analysis

Statistical analyses followed the intention-to-treat principle. Mixed regression models were applied to account for missing data and the nested structure of the data (ie, repeated observations within patients, patients within subgroups, subgroups within treatment groups, and treatment groups within institutions). Most continuous outcomes were skewed and analyzed with generalized linear mixed-model (GLMM) gamma regression with a log link. Alternative distributions were used where appropriate. Treatment retention was analyzed using GLMM survival analysis with dropout per quarter as the outcome.

For the repeated measures part of the model, 8 different covariance structures were tested. The fixed part of the model contained time, treatment, and their interaction, with time transformed by either a linear, quadratic, or square root function. Random effects for time and intercept and, where applicable, the treatment $\times$ time interaction were evaluated at multiple hierarchical levels and included only when this significantly improved model fit. For the final model, estimated means with 95% CIs were derived at 6-month intervals over a 3-year period. Between-group and within-group Cohen d effect sizes were calculated, as well as r values for fixed effects. Model assumptions were checked using residual analysis. Finally, robustness analyses and sensitivity analyses were performed. Robustness analyses included an autoregressive moving average (1,1) covariance structure³⁸ and random effects where estimation permitted. Four sensitivity analyses were conducted for the primary outcome to assess the impact of (1) COVID-19-related treatment interruption, (2) protocol deviations, (3) dropout, and (4) psychotropic medication use.

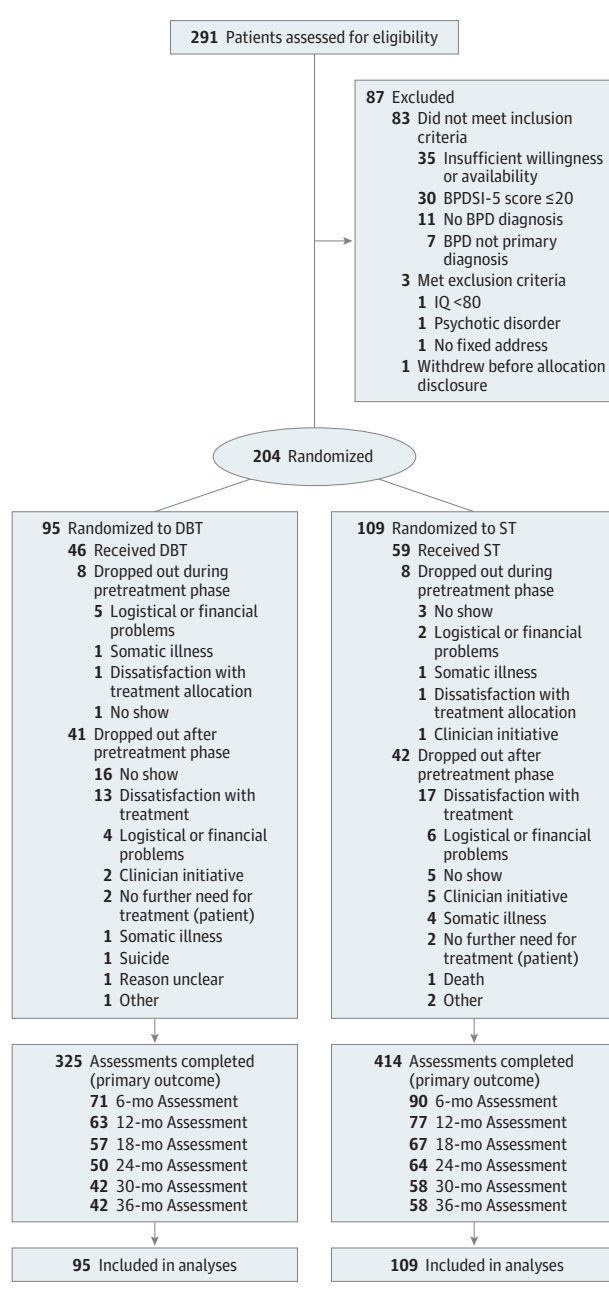
Analyses were conducted using SAS software, version 9.4 (SAS Institute). Statistical significance was defined as a 2-sided $P < .05$. The statistical analyses are described in detail in the eMethods in [Supplement 2](#).

Results

Sample Characteristics and Study Flow

Of 291 patients referred to the study, 87 were not eligible for participation, yielding a final sample of 204 participants (172 female [84.3%]; 30 male [14.7%]; 2 other [1.0%]; mean [SD] age, 32.21 [9.57] years). Among them, 95 (46.6%) received DBT and 109 (53.4%) received ST. The CONSORT diagram is presented in [Figure 1](#). Demographic and clinical characteristics of the sample are reported in [Table 1](#).

Figure 1. CONSORT Flowchart of Participants



Treatment Outcomes

Primary Outcome

At baseline, the estimated mean BPDSI-5 scores were 30.56 (95% CI, 28.02-33.34) for DBT and 29.70 (95% CI, 27.30-32.31) for ST. At 3 years, scores were 7.99 (95% CI, 6.44-9.91) for DBT and 9.08 (95% CI, 7.47-11.03) for ST. No significant treatment-by-time interaction emerged ($P = .27$, $r = 0.09$), indicating no significant difference between DBT and ST in reducing total BPDSI-5 scores. The model-estimated mean follow-up scores at 36 months did not significantly differ either ($P = .36$; transformed scale: $\Delta M_{\text{DBT-ST}} = -0.13$; 95% CI, -0.40 to 0.15 ; original BPDSI-5 scale: $\Delta M_{\text{DBT-ST}} = -1.09$; $d = 0.15$; 95%

Table 1. Demographic and Clinical Characteristics at Baseline

Characteristic	Participants, No. (%)	
	DBT (n = 95)	ST (n = 109)
Age, mean (SD), y	31.93 (9.35)	32.45 (9.80)
Educational level, mean (SD) ^a	4.32 (1.64)	4.09 (1.66)
BPD severity, mean (SD) ^b	31.27 (7.08)	30.25 (7.50)
Comorbid disorders, mean (SD), No.		
DSM-5 syndrome disorders ^c	2.94 (1.70)	2.57 (1.67)
DSM-5 personality disorders ^d	0.52 (0.73)	0.51 (0.78)
Sex		
Female	82 (86)	90 (83)
Male	12 (13)	18 (17)
Other ^e	1 (1)	1 (1)
Dutch ethnicity	69 (73)	81 (74)
Employment status		
Employed	30 (32)	24 (22)
Employed, sick leave	17 (18)	25 (23)
Work disabled	21 (22)	21 (19)
Welfare	8 (8)	11 (10)
Unemployed	2 (2)	3 (3)
Student	15 (16)	25 (23)
Other	2 (2)	0
Comorbid DSM-5 personality disorder		
Cluster A	5 (5)	12 (11)
Cluster B	4 (4)	7 (6)
Cluster C	34 (36)	29 (27)
None	57 (60)	70 (64)
Comorbid DSM-5 syndrome disorder		
Affective disorders	61 (64)	62 (57)
Anxiety disorders	45 (47)	43 (39)
Substance use disorder	32 (34)	37 (34)
Attention-deficit/hyperactivity disorder	32 (34)	30 (28)
Posttraumatic stress disorder	22 (23)	24 (22)
Sleep disorders	20 (21)	22 (20)
Eating disorders	11 (12)	15 (14)
Other	20 (21)	19 (17)
None	6 (6)	12 (11)
Psychotropic medication use	62 (65)	69 (63)

Abbreviations: BPD, borderline personality disorder; BPDSI-5, Borderline Personality Disorder Severity Index, fifth version; DSM-5, *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition*; SCID-5-PD, Structured Clinical Interview for DSM-5 Personality Disorders; SCID-5-S, Structured Clinical Interview for DSM-5 Syndrome Disorders.

^a Based on the International Standard Classification of Education, 2011 version.

^b Based on the BPDSI-5.

^c Based on the SCID-5-S.

^d Based on the SCID-5-PD (excluding a BPD diagnosis).

^e Includes nonbinary and gender questioning.

CI, -0.17 to 0.47). A significant overall reduction in BPD severity over time was observed, with large within-group effect sizes over the 3-year period (pre-follow-up) in both conditions (Table 2, Table 3, and Figure 2A). Notably, in both conditions, the estimated mean total BPDSI-5 score was below the

Table 2. Results of the Mixed Regression Analyses

Outcome, effect ^a	b (SE)	t	df ^b	P value	r
Primary outcome					
BPDSI-5 total score					
Time	-0.0081 (0.0005)	-17.94	135	<.001	.84
Time × treatment	0.0010 (0.0009)	1.11	136	.27	.09
Secondary outcomes: BPDSI-5 subscales^c					
1: Abandonment					
Time	-0.0114 (0.0008)	-13.86	332	<.001	.61
Time × treatment	0.0030 (0.0016)	1.83	332	.07	.10
2: Interpersonal relationships					
Time	-0.0087 (0.0006)	-15.21	253	<.001	.69
Time × treatment	-0.0004 (0.0011)	-0.33	253	.74	.02
3: Identity					
Time	-0.0117 (0.0008)	-14.03	146	<.001	.76
Time × treatment	0.0017 (0.0017)	1.03	146	.30	.08
4: Impulsivity					
Time	-0.0077 (0.0007)	-11.84	207	<.001	.64
Time × treatment	-0.0011 (0.0013)	-0.82	207	.11	.06
5: (Para)suicide					
Time	-0.0124 (0.0026)	-4.79	5	.004	.90
Time × treatment	0.0039 (0.0035)	1.11	11	.29	.31
6: Affective instability					
Time	-0.0056 (0.0004)	-12.80	141	<.001	.73
Time × treatment	0.0011 (0.0009)	1.27	141	.21	.11
7: Emptiness					
Time	-0.0059 (0.0005)	-11.16	118	<.001	.72
Time × Treatment	-0.0004 (0.0010)	-0.40	118	.69	.04
8: Anger					
Time	-0.0111 (0.0008)	-13.26	227	<.001	.66
Time × treatment	0.0028 (0.0017)	1.69	227	.09	.11
9: Dissociation or paranoia					
Time	-0.0099 (0.0010)	-10.20	277	<.001	.52
Time × treatment	0.0023 (0.0019)	1.21	277	.23	.07
Other secondary outcomes					
uBPDc					
Time	-0.0036 (0.0002)	-16.66	124	<.001	.83
Time × treatment	0.0005 (0.0004)	1.05	124	.30	.09
BSI					
Time	-0.0045 (0.0004)	-11.02	134	<.001	.69
Time × treatment	0.0010 (0.0008)	1.22	134	.23	.10
WHODAS 2.0					
Time	-0.0047 (0.0004)	-10.59	137	<.001	.67
Time × treatment	-0.0005 (0.0009)	-0.52	137	.60	.04
EQ-5D-5L ^d					
Time	-0.0036 (0.0004)	-8.46	131	<.001	.60
Time × treatment	-0.0002 (0.0009)	-0.17	131	.86	.01
MHQoL-7D ^d					
Time	-0.0064 (0.0007)	-9.34	153	<.001	.60
Time × treatment	0.0012 (0.0014)	0.90	153	.37	.07
Happiness					
Time ^e	0.1280 (0.0108)	11.87	37	<.001	.89
Time ^e × treatment	-0.0209 (0.0216)	-0.97	37	.34	.16

(continued)

Table 2. Results of the Mixed Regression Analyses (continued)

Outcome, effect ^a	b (SE)	t	df ^b	P value	r
ORS					
Time ^e	1.8502 (0.1372)	13.47	143	<.001	.75
Time ^e × treatment	−0.2509 (0.2747)	−0.91	143	.36	.08
ISI					
Time	−0.0023 (0.0003)	−6.57	260	<.001	.38
Time × treatment	0.0005 (0.0007)	0.77	260	.44	.05
NFQ ^f					
Time	−0.0036 (0.0008)	−4.78	158	<.001	.36
Time × treatment	0.0001 (0.0015)	0.05	158	.96	.00
Work or study status					
Time	0.0046 (0.0016)	2.80	741	.005	.10
Time × treatment	−0.0011 (0.0025)	−0.42	741	.67	.02

Abbreviations: BPDSI-5, Borderline Personality Disorder Severity Index, fifth version; BSI, Brief Symptom Inventory; DBT, dialectical behavior therapy; EQ-5D-5L, 5-level EuroQol 5-dimensional version; ISI, Insomnia Severity Index; MHQoL-7D, Mental Health Quality of Life 7-dimensional Questionnaire; NFQ, Nightmare Frequency Questionnaire; ORS, Outcome Rating Scale; ST, schema therapy; uBPDc, ultrashort BPD checklist; WHODAS 2.0, World Health Organization Disability Assessment Schedule 2.0.

^a Time was modeled as a linear function unless specified otherwise. The time coefficient represents the overall slope across treatment conditions.

^b Denominator degrees of freedom, presented in rounded values.

^c To enable negative binomial regression, sum scores were calculated for

each subscale.

^d To enable gamma regression, the distribution was mirrored.

^e Square root of time.

^f A significant main effect for treatment was found. To account for baseline differences, the analysis was repeated with the baseline NFQ score included as a covariate and the baseline scores excluded from the dependent time variable. Results were consistent, showing a significant time effect, no significant interaction effect and, based on contrast analysis, no significant difference between the adjusted estimated means at the follow-up assessment (3 years).

clinical cutoff score of 15 at posttreatment (2 years after baseline), indicating recovery (Table 3). The robustness and sensitivity analyses yielded similar findings (eTables 1 and 2 in Supplement 2).

Secondary Outcomes

Treatment Retention The overall dropout rate was 31% at 1 year and 49% at 2 years (DBT: 34% and 52%, respectively; ST: 29% and 46%, respectively). Figure 2B displays the estimated treatment retention over the course of the treatment (<0 months: pretreatment phase, 0: start of the treatment phase). GLMM survival analysis revealed no significant differences in dropout rates between the treatments ($F_{8,1354} = 0.35$, $P = .94$).

Severity per BPD Domain The GLMM analyses yielded no significant treatment-by-time interactions, indicating that DBT and ST did not differ significantly in reductions in *DSM-5* BPD criteria. Between-group differences at follow-up were small. Significant main effects of time for all BPDSI-5 subscales were found, with substantial reductions in both treatment conditions (Table 2 and Table 3).

Other Secondary Outcomes For all 10 secondary outcomes, no significant treatment-by-time interactions emerged; in other words, change trajectories did not differ significantly between DBT and ST on the secondary outcomes. Likewise, follow-up between-group differences were small. Significant main effects of time were found, reflecting overall improvement over time with large within-group pre-follow-up effect sizes in both treatments, except for work or study status and sleep problems (Table 2 and Table 3).

Furthermore, 2 patients died during the study period. One participant in the DBT group died by suicide in the sixth quarter (months 15-18 after the start of the treatment phase), and 1 participant in the ST group passed away due to a somatic cause in the first quarter (months 0-3 after the start of the treatment phase). Additionally, 7 other SAEs were reported among 6 participants (ST: $n = 5$; DBT: $n = 1$), including 6 suicide attempts and 1 inpatient admission due to suicidal ideation. Six events occurred during treatment and 1 after the participant had dropped out.

Finally, robustness analyses for the secondary outcomes yielded similar findings, except for the uBPDc, where DBT outperformed ST ($P = .04$, $r = 0.14$), although the difference at 36-month follow-up was small ($d = 0.31$) (eTables 3 and 4 in Supplement 2). Specifications of the statistical analyses, including robustness analyses, are provided in eTables 1-3 in Supplement 2.

Discussion

This study compared the effectiveness of DBT and ST for outpatients with BPD in the context of a multicenter RCT. Treatments were delivered over 2 years in a combined group-individual format. No significant differential treatment effects emerged for BPD severity or any of the secondary outcomes, including *DSM-5* BPD criteria, general symptom severity, functioning, quality of life, well-being, and sleep problems. Both treatments demonstrated improvements across all outcomes over the 3-year study period. Treatment retention also did not differ significantly, with dropout rates of 34% for DBT and 29% for ST at 1 year and 52% for DBT and 46% for ST after 2 years.

Table 3. Primary and Secondary Outcomes Across all Time Points

Outcome by time, y ^a	DBT		ST		
	Estimated mean (95% CI) ^b	Within-group effect size, Cohen d ^c	Estimated mean (95% CI) ^b	Within-group effect size, Cohen d ^c	Between-group effect size, Cohen d (95% CI) ^c
Primary outcome					
BPDSI-5 total score					
0	30.56 (28.02 to 33.34)	NA	29.70 (27.30 to 32.31)	NA	-0.12 (-0.38 to 0.15)
0.5	24.44 (22.34 to 26.74)	0.84	24.38 (22.36 to 26.58)	0.73	-0.01 (-0.25 to 0.24)
1	19.54 (17.62 to 21.68)	1.51	20.01 (18.15 to 22.05)	1.33	0.06 (-0.22 to 0.35)
1.5	15.63 (13.77 to 17.73)	2.05	16.42 (14.62 to 18.45)	1.82	0.11 (-0.21 to 0.43)
2	12.49 (10.71 to 14.57)	2.47	13.48 (11.71 to 15.51)	2.22	0.13 (-0.20 to 0.47)
2.5	9.99 (8.31 to 12.01)	2.82	11.06 (9.36 to 13.07)	2.55	0.15 (-0.19 to 0.48)
3	7.99 (6.44 to 9.91)	3.09	9.08 (7.47 to 11.03)	2.82	0.15 (-0.17 to 0.47)
Secondary outcomes: BPDSI-5 subscales^d					
1: Abandonment					
0	21.14 (18.77 to 23.81)	NA	20.72 (18.54 to 23.15)	NA	-0.03 (-0.31 to 0.24)
0.5	15.11 (13.33 to 17.12)	0.48	16.01 (14.33 to 17.88)	0.38	0.07 (-0.14 to 0.28)
1	10.79 (9.19 to 12.68)	0.83	12.37 (10.83 to 14.13)	0.67	0.13 (-0.07 to 0.32)
1.5	7.71 (6.24 to 9.53)	1.08	9.56 (8.07 to 11.32)	0.90	0.15 (-0.04 to 0.34)
2	5.51 (4.21 to 7.21)	1.25	7.39 (5.98 to 9.13)	1.07	0.15 (-0.03 to 0.33)
2.5	3.94 (2.83 to 5.48)	1.38	5.71 (4.41 to 7.39)	1.20	0.14 (-0.02 to 0.30)
3	2.81 (1.90 to 4.16)	1.47	4.41 (3.25 to 6.00)	1.31	0.13 (-0.01 to 0.27)
2: Interpersonal relationships					
0	24.12 (21.92 to 26.53)	NA	24.68 (22.61 to 26.95)	NA	0.05 (-0.22 to 0.32)
0.5	19.31 (17.64 to 21.14)	0.41	19.57 (18.01 to 21.27)	0.44	0.02 (-0.18 to 0.23)
1	15.46 (13.91 to 17.19)	0.75	15.52 (14.10 to 17.08)	0.79	0.00 (-0.19 to 0.19)
1.5	12.38 (10.83 to 14.16)	1.01	12.30 (10.91 to 13.88)	1.07	-0.01 (-0.20 to 0.18)
2	9.92 (8.37 to 11.74)	1.22	9.76 (8.39 to 11.35)	1.29	-0.01 (-0.21 to 0.18)
2.5	7.94 (6.45 to 9.77)	1.39	7.74 (6.43 to 9.31)	1.46	-0.02 (-0.21 to 0.17)
3	6.36 (4.96 to 8.14)	1.53	6.13 (4.92 to 7.65)	1.60	-0.02 (-0.20 to 0.16)
3: Identity					
0	13.27 (12.21 to 14.42)	NA	13.60 (12.61 to 14.67)	NA	0.06 (-0.20 to 0.32)
0.5	9.57 (8.71 to 10.51)	0.63	10.26 (9.45 to 11.14)	0.57	0.12 (-0.09 to 0.33)
1	6.90 (6.01 to 7.94)	1.09	7.74 (6.88 to 8.70)	1.00	0.14 (-0.08 to 0.37)
1.5	4.98 (4.09 to 6.06)	1.42	5.84 (4.96 to 6.88)	1.33	0.15 (-0.09 to 0.38)
2	3.59 (2.77 to 4.65)	1.66	4.40 (3.56 to 5.45)	1.57	0.14 (-0.09 to 0.37)
2.5	2.59 (1.88 to 3.57)	1.83	3.32 (2.55 to 4.33)	1.76	0.13 (-0.08 to 0.33)
3	1.87 (1.27 to 2.75)	1.95	2.51 (1.82 to 3.45)	1.90	0.11 (-0.08 to 0.29)
4: Impulsivity					
0	15.24 (13.18 to 17.63)	NA	16.29 (14.29 to 18.57)	NA	0.09 (-0.18 to 0.36)
0.5	12.65 (10.97 to 14.59)	0.19	13.15 (11.56 to 14.96)	0.24	0.04 (-0.17 to 0.26)
1	10.50 (8.99 to 12.26)	0.37	10.61 (9.21 to 12.24)	0.45	0.01 (-0.19 to 0.21)
1.5	8.71 (7.27 to 10.45)	0.52	8.57 (7.25 to 10.12)	0.62	-0.01 (-0.20 to 0.17)
2	7.23 (5.83 to 8.97)	0.64	6.92 (5.67 to 8.44)	0.76	-0.03 (-0.21 to 0.15)
2.5	6.00 (4.65 to 7.74)	0.75	5.58 (4.41 to 7.06)	0.87	-0.04 (-0.21 to 0.14)
3	4.98 (3.70 to 6.70)	0.83	4.51 (3.43 to 5.92)	0.95	-0.04 (-0.21 to 0.13)

(continued)

Table 3. Primary and Secondary Outcomes Across all Time Points (continued)

Outcome by time, y ^a	DBT		ST		
	Estimated mean (95% CI) ^b	Within-group effect size, Cohen d ^c	Estimated mean (95% CI) ^b	Within-group effect size, Cohen d ^c	Between-group effect size, Cohen d (95% CI) ^c
5: (Para)suicide					
0	9.53 (7.28 to 12.49)	NA	9.65 (7.52 to 12.37)	NA	0.01 (−0.30 to 0.32)
0.5	6.57 (5.06 to 8.53)	0.26	7.35 (5.77 to 9.37)	0.20	0.07 (−0.13 to 0.27)
1	4.53 (3.09 to 6.63)	0.45	5.61 (3.92 to 8.02)	0.36	0.10 (−0.09 to 0.28)
1.5	3.12 (1.79 to 5.44)	0.57	4.28 (2.52 to 7.24)	0.48	0.10 (−0.09 to 0.29)
2	2.15 (1.03 to 4.50)	0.66	3.26 (1.61 to 6.58)	0.57	0.10 (−0.08 to 0.28)
2.5	1.48 (0.59 to 3.74)	0.72	2.49 (1.03 to 5.98)	0.64	0.09 (−0.08 to 0.26)
3	1.02 (0.34 to 3.10)	0.76	1.90 (0.66 to 5.44)	0.69	0.08 (−0.07 to 0.22)
6: Affective instability					
0	35.31 (33.80 to 36.89)	NA	34.27 (32.89 to 35.71)	NA	−0.13 (−0.39 to 0.13)
0.5	30.12 (28.60 to 31.72)	0.65	30.08 (28.71 to 31.52)	0.52	−0.00 (−0.27 to 0.26)
1	25.69 (23.81 to 27.72)	1.21	26.41 (24.72 to 28.21)	0.99	0.09 (−0.24 to 0.42)
1.5	21.92 (19.72 to 24.36)	1.68	23.18 (21.17 to 25.38)	1.39	0.16 (−0.24 to 0.55)
2	18.70 (16.29 to 21.45)	2.08	20.35 (18.10 to 22.88)	1.75	0.21 (−0.23 to 0.65)
2.5	15.95 (13.45 to 18.91)	2.43	17.86 (15.46 to 20.64)	2.06	0.24 (−0.23 to 0.71)
3	13.60 (11.10 to 16.67)	2.72	15.68 (13.20 to 18.63)	2.33	0.26 (−0.23 to 0.75)
7: Emptiness					
0	24.18 (22.48 to 26.01)	NA	23.75 (22.19 to 25.42)	NA	−0.05 (−0.30 to 0.21)
0.5	20.87 (19.32 to 22.56)	0.35	20.28 (18.86 to 21.80)	0.37	−0.06 (−0.29 to 0.17)
1	18.01 (16.31 to 19.90)	0.65	17.31 (15.78 to 18.98)	0.68	−0.07 (−0.33 to 0.18)
1.5	15.55 (13.65 to 17.71)	0.91	14.78 (13.10 to 16.66)	0.95	−0.08 (−0.37 to 0.20)
2	13.42 (11.38 to 15.83)	1.14	12.61 (10.83 to 14.68)	1.18	−0.09 (−0.39 to 0.22)
2.5	11.58 (9.46 to 14.17)	1.33	10.77 (8.94 to 12.96)	1.37	−0.09 (−0.41 to 0.24)
3	10.00 (7.86 to 12.70)	1.50	9.19 (7.38 to 11.45)	1.54	−0.09 (−0.42 to 0.25)
8: Anger					
0	14.03 (11.98 to 16.42)	NA	13.95 (12.06 to 16.14)	NA	−0.01 (−0.27 to 0.26)
0.5	10.14 (8.7 to 11.81)	0.34	10.85 (9.43 to 12.48)	0.27	0.06 (−0.13 to 0.25)
1	7.33 (6.16 to 8.71)	0.59	8.44 (7.23 to 9.86)	0.48	0.10 (−0.06 to 0.26)
1.5	5.29 (4.28 to 6.55)	0.77	6.57 (5.45 to 7.91)	0.65	0.11 (−0.04 to 0.26)
2	3.83 (2.94 to 4.98)	0.90	5.11 (4.07 to 6.41)	0.78	0.11 (−0.02 to 0.25)
2.5	2.77 (2.01 to 3.80)	0.99	3.97 (3.02 to 5.22)	0.88	0.11 (−0.02 to 0.23)
3	2.00 (1.37 to 2.91)	1.06	3.09 (2.24 to 4.27)	0.95	0.10 (−0.01 to 0.20)
9: Dissociation or paranoia					
0	13.23 (10.91 to 16.04)	NA	13.09 (10.94 to 15.65)	NA	−0.01 (−0.29 to 0.27)
0.5	9.93 (8.23 to 11.97)	0.27	10.44 (8.82 to 12.36)	0.28	0.04 (−0.17 to 0.25)
1	7.45 (6.02 to 9.21)	0.50	8.32 (6.95 to 9.97)	0.52	0.07 (−0.11 to 0.25)
1.5	5.59 (4.31 to 7.24)	0.70	6.64 (5.37 to 8.19)	0.73	0.08 (−0.08 to 0.25)
2	4.19 (3.05 to 5.76)	0.87	5.29 (4.11 to 6.81)	0.90	0.09 (−0.07 to 0.24)
2.5	3.15 (2.14 to 4.62)	1.02	4.22 (3.12 to 5.71)	1.05	0.09 (−0.06 to 0.23)
3	2.36 (1.50 to 3.71)	1.15	3.37 (2.36 to 4.80)	1.18	0.08 (−0.05 to 0.21)

(continued)

No significant differences between DBT and ST in the rates of change were observed for any of the outcome measures. The high similarity in outcome trajectories over time is remark-

able given the differences between the treatments in conceptualization and etiological models of BPD as well as therapeutic techniques, despite their shared cognitive-behavioral roots.

Table 3. Primary and Secondary Outcomes Across all Time Points (continued)

Outcome by time, y ^a	DBT		ST		
	Estimated mean (95% CI) ^b	Within-group effect size, Cohen d ^c	Estimated mean (95% CI) ^b	Within-group effect size, Cohen d ^c	Between-group effect size, Cohen d (95% CI) ^c
Other secondary outcomes					
uBPDc					
0	29.32 (28.10 to 30.59)	NA	29.39 (28.26 to 30.57)	NA	0.01 (−0.24 to 0.27)
0.5	26.56 (25.46 to 27.70)	0.41	26.94 (25.91 to 28.00)	0.37	0.06 (−0.17 to 0.29)
1	24.05 (22.93 to 25.23)	0.79	24.69 (23.64 to 25.78)	0.71	0.10 (−0.14 to 0.33)
1.5	21.79 (20.56 to 23.09)	1.13	22.63 (21.48 to 23.83)	1.02	0.13 (−0.13 to 0.39)
2	19.73 (18.39 to 21.18)	1.44	20.74 (19.48 to 22.08)	1.30	0.15 (−0.14 to 0.44)
2.5	17.88 (16.42 to 19.45)	1.72	19.01 (17.64 to 20.48)	1.56	0.17 (−0.14 to 0.48)
3	16.19 (14.66 to 17.89)	1.97	17.42 (15.96 to 19.01)	1.80	0.18 (−0.15 to 0.52)
BSI					
0	1.63 (1.51 to 1.77)	NA	1.57 (1.46 to 1.69)	NA	−0.10 (−0.38 to 0.17)
0.5	1.44 (1.33 to 1.56)	0.31	1.41 (1.32 to 1.52)	0.24	−0.04 (−0.28 to 0.21)
1	1.26 (1.15 to 1.38)	0.59	1.28 (1.17 to 1.39)	0.46	0.02 (−0.23 to 0.27)
1.5	1.11 (0.99 to 1.24)	0.83	1.15 (1.04 to 1.27)	0.66	0.06 (−0.20 to 0.33)
2	0.98 (0.85 to 1.12)	1.04	1.04 (0.92 to 1.17)	0.84	0.10 (−0.19 to 0.39)
2.5	0.86 (0.73 to 1.01)	1.23	0.94 (0.81 to 1.08)	1.00	0.12 (−0.18 to 0.43)
3	0.76 (0.63 to 0.91)	1.39	0.85 (0.72 to 1.00)	1.15	0.14 (−0.18 to 0.46)
WHODAS 2.0					
0	1.53 (1.41 to 1.65)	NA	1.54 (1.43 to 1.65)	NA	0.02 (−0.24 to 0.27)
0.5	1.36 (1.26 to 1.46)	0.27	1.35 (1.26 to 1.45)	0.30	−0.01 (−0.24 to 0.21)
1	1.21 (1.10 to 1.32)	0.51	1.19 (1.09 to 1.29)	0.56	−0.03 (−0.27 to 0.20)
1.5	1.07 (0.96 to 1.20)	0.72	1.04 (0.94 to 1.16)	0.79	−0.05 (−0.31 to 0.21)
2	0.95 (0.83 to 1.10)	0.92	0.92 (0.80 to 1.04)	0.99	−0.06 (−0.35 to 0.23)
2.5	0.85 (0.71 to 1.01)	1.08	0.80 (0.69 to 0.94)	1.17	−0.07 (−0.38 to 0.24)
3	0.75 (0.61 to 0.93)	1.24	0.71 (0.59 to 0.85)	1.33	−0.08 (−0.40 to 0.25)
EQ-5D-5L					
0	0.56 (0.52 to 0.60)	NA	0.54 (0.50 to 0.58)	NA	0.08 (−0.15 to 0.31)
0.5	0.60 (0.56 to 0.63)	0.16	0.58 (0.54 to 0.62)	0.17	0.07 (−0.14 to 0.28)
1	0.63 (0.59 to 0.67)	0.30	0.62 (0.58 to 0.66)	0.33	0.06 (−0.16 to 0.27)
1.5	0.67 (0.62 to 0.71)	0.44	0.66 (0.61 to 0.69)	0.47	0.05 (−0.18 to 0.28)
2	0.70 (0.65 to 0.74)	0.56	0.69 (0.64 to 0.73)	0.60	0.04 (−0.22 to 0.29)
2.5	0.73 (0.67 to 0.77)	0.66	0.72 (0.67 to 0.76)	0.72	0.03 (−0.24 to 0.30)
3	0.75 (0.69 to 0.80)	0.77	0.74 (0.69 to 0.79)	0.82	0.02 (−0.26 to 0.31)
MHQoL-7D					
0	0.29 (0.21 to 0.37)	NA	0.30 (0.22 to 0.36)	NA	−0.00 (−0.29 to 0.28)
0.5	0.41 (0.35 to 0.47)	0.34	0.40 (0.34 to 0.45)	0.18	0.05 (−0.18 to 0.28)
1	0.51 (0.45 to 0.57)	0.63	0.48 (0.42 to 0.53)	0.34	0.09 (−0.14 to 0.32)
1.5	0.60 (0.53 to 0.65)	0.86	0.56 (0.49 to 0.61)	0.49	0.12 (−0.13 to 0.37)
2	0.67 (0.59 to 0.73)	1.06	0.62 (0.55 to 0.68)	0.62	0.13 (−0.14 to 0.41)
2.5	0.72 (0.64 to 0.78)	1.22	0.67 (0.59 to 0.74)	0.74	0.14 (−0.15 to 0.43)
3	0.77 (0.69 to 0.83)	1.36	0.72 (0.64 to 0.79)	0.86	0.14 (−0.15 to 0.43)

(continued)

Nonetheless, our findings align with the overall findings of the previous DBT-ST trial,¹⁴ which suggested the importance of shared factors (eg, explicit framework, focus on emotions). Ad-

ditionally, different therapeutic techniques may lead to comparable outcomes through different mechanisms. However, the absence of statistically significant treatment differences should

Table 3. Primary and Secondary Outcomes Across all Time Points (continued)

Outcome by time, y ^a	DBT		ST		
	Estimated mean (95% CI) ^b	Within-group effect size, Cohen d ^c	Estimated mean (95% CI) ^b	Within-group effect size, Cohen d ^c	Between-group effect size, Cohen d (95% CI) ^c
Happiness					
0	3.19 (3.00 to 3.38)	NA	3.18 (3.01 to 3.36)	NA	0.00 (−0.25 to 0.26)
0.5	3.89 (3.70 to 4.08)	0.69	3.78 (3.61 to 3.96)	0.58	0.11 (−0.14 to 0.36)
1	4.18 (3.96 to 4.41)	0.97	4.03 (3.82 to 4.24)	0.82	0.15 (−0.15 to 0.45)
1.5	4.41 (4.14 to 4.68)	1.19	4.22 (3.98 to 4.46)	1.01	0.18 (−0.17 to 0.54)
2	4.60 (4.29 to 4.90)	1.37	4.38 (4.10 to 4.66)	1.17	0.21 (−0.19 to 0.61)
2.5	4.77 (4.43 to 5.10)	1.54	4.52 (4.22 to 4.83)	1.30	0.24 (−0.21 to 0.68)
3	4.92 (4.55 to 5.28)	1.68	4.65 (4.32 to 4.99)	1.43	0.26 (−0.23 to 0.74)
ORS					
0	43.33 (39.94 to 46.71)	NA	43.33 (40.19 to 46.47)	NA	−0.00 (−0.26 to 0.26)
0.5	53.40 (50.32 to 56.48)	0.57	52.12 (49.30 to 54.95)	0.50	0.07 (−0.16 to 0.31)
1	57.57 (54.23 to 60.92)	0.81	55.77 (52.73 to 58.80)	0.70	0.10 (−0.15 to 0.36)
1.5	60.77 (57.10 to 64.45)	0.99	58.56 (55.26 to 61.87)	0.86	0.13 (−0.15 to 0.40)
2	63.47 (59.46 to 67.49)	1.14	60.92 (57.33 to 64.51)	1.00	0.14 (−0.16 to 0.45)
2.5	65.85 (61.50 to 70.20)	1.27	62.99 (59.12 to 66.87)	1.11	0.16 (−0.17 to 0.49)
3	68.00 (63.32 to 72.68)	1.40	64.87 (60.71 to 69.03)	1.22	0.18 (−0.18 to 0.53)
ISI					
0	14.37 (12.94 to 15.96)	NA	14.65 (13.29 to 16.15)	NA	0.04 (−0.28 to 0.36)
0.5	13.44 (12.21 to 14.79)	0.14	13.90 (12.71 to 15.19)	0.12	0.07 (−0.21 to 0.35)
1	12.57 (11.45 to 13.81)	0.28	13.18 (12.09 to 14.37)	0.23	0.09 (−0.16 to 0.35)
1.5	11.76 (10.65 to 12.99)	0.40	12.51 (11.43 to 13.69)	0.33	0.11 (−0.14 to 0.37)
2	11.01 (9.85 to 12.29)	0.52	11.86 (10.74 to 13.11)	0.43	0.13 (−0.13 to 0.40)
2.5	10.30 (9.07 to 11.69)	0.63	11.26 (10.05 to 12.60)	0.52	0.15 (−0.14 to 0.43)
3	9.63 (8.32 to 11.15)	0.73	10.68 (9.38 to 12.16)	0.61	0.16 (−0.15 to 0.47)
NFQ ^e					
0	3.54 (2.84 to 4.41)	NA	4.70 (3.94 to 5.60)	NA	0.26 (0.00 to 0.52)
0.5	3.22 (2.62 to 3.95)	0.07	4.28 (3.63 to 5.05)	0.09	0.24 (0.02 to 0.46)
1	2.93 (2.38 to 3.61)	0.14	3.90 (3.31 to 4.60)	0.18	0.22 (0.02 to 0.42)
1.5	2.66 (2.12 to 3.35)	0.20	3.56 (2.97 to 4.25)	0.26	0.20 (0.00 to 0.40)
2	2.42 (1.87 to 3.15)	0.25	3.24 (2.65 to 3.97)	0.33	0.18 (−0.03 to 0.39)
2.5	2.20 (1.63 to 2.99)	0.30	2.95 (2.34 to 3.73)	0.39	0.17 (−0.06 to 0.43)
3	2.01 (1.41 to 2.85)	0.35	2.69 (2.06 to 3.52)	0.45	0.16 (−0.08 to 0.39)
Work or study status					
0	0.56 (0.46 to 0.66)	NA	0.52 (0.43 to 0.61)	NA	0.09 (−0.19 to 0.36)
0.5	0.58 (0.49 to 0.67)	0.05	0.55 (0.46 to 0.63)	0.06	0.08 (−0.23 to 0.38)
1	0.61 (0.52 to 0.69)	0.09	0.58 (0.50 to 0.65)	0.12	0.06 (−0.27 to 0.39)
1.5	0.63 (0.54 to 0.71)	0.13	0.61 (0.52 to 0.68)	0.18	0.05 (−0.30 to 0.40)
2	0.65 (0.55 to 0.74)	0.18	0.63 (0.54 to 0.71)	0.23	0.03 (−0.34 to 0.41)
2.5	0.67 (0.56 to 0.76)	0.22	0.66 (0.56 to 0.75)	0.29	0.02 (−0.37 to 0.41)
3	0.69 (0.57 to 0.79)	0.26	0.69 (0.58 to 0.78)	0.35	0.01 (−0.39 to 0.41)

Abbreviations: BPD5I-5, Borderline Personality Disorder Severity Index, fifth version; BSI, Brief Symptom Inventory; DBT, dialectical behavior therapy; EQ-5D-5L, 5-level EuroQol 5-dimensional version; ISI, Insomnia Severity Index; MHQoL-7D, Mental Health Quality of Life 7-dimensional Questionnaire; NA, not applicable; NFQ, Nightmare Frequency Questionnaire; ORS, Outcome Rating Scale; ST, schema therapy; uBPDc, Ultrashort BPD Checklist; WHODAS 2.0, World Health Organization Disability Assessment Schedule 2.0.

^a Time in years.

^b Estimated means and 95% CIs are in original scale.

^c Effect sizes were calculated using the observed baseline standard deviation for the denominator. For work or study status, Cohen *h* was calculated, based on the estimated proportions. Positive within-group effect sizes indicate a positive effect. Positive between-group effect sizes indicate an effect in favor of DBT.

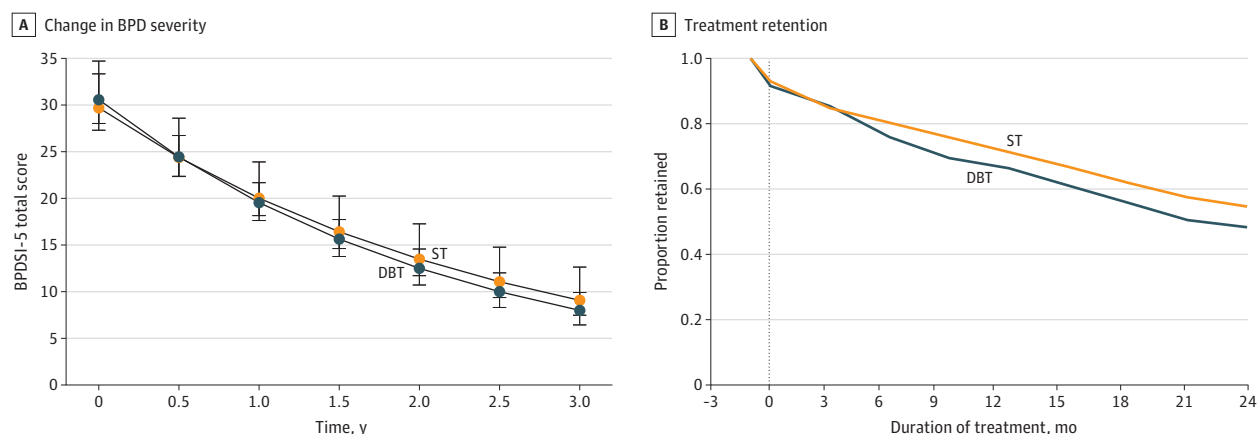
^d The estimates represent sum scores for each subscale.

^e Not adjusted for baseline scores.

not be interpreted as evidence of equivalence between DBT and ST, as the study was not designed or powered as a noninferiority or equivalence trial. The trial was powered to detect medium-sized treatment effects, and because both DBT and ST are evidence-based treatments, differential treatment ef-

fects may be small and may therefore have remained undetected. Moreover, the absence of significant average treatment differences does not imply that individual patients respond similarly, as heterogeneous treatment effects are common in BPD.^{39,40} Personalized treatment selection proce-

Figure 2. Line Graphs of Treatment Effects on Borderline Personality Disorder (BPD) Severity and Treatment Retention



BPDSI-5 indicates Borderline Personality Disorder Severity Index, fifth version; DBT, dialectical behavior therapy; ST, schema therapy.

dures using clinical prediction models may therefore be promising⁴¹ and are part of future work in the BOOTS project.¹⁹ Additionally, combining multiple BPD trials using individual participant data meta-analysis is a promising approach to develop personalized treatments while addressing the limited power of individual trials.⁴²

While differential treatment effects were not found, both treatments showed substantial improvements across outcomes, generally with large effect sizes, consistent with the other DBT-ST trial.¹⁴ However, effect sizes for participation in work or study and sleep problems were smaller, aligning with meta-analysis findings demonstrating that improvements in functioning often fall below thresholds for clinical relevance.¹⁷ In addition, sleep problems receive little attention in clinical guidelines and BPD treatment, despite their impact. Integrating sleep-focused interventions may therefore be beneficial.⁴³ However, it is important to note that the study did not include a no-treatment or nonspecialized control group, limiting conclusions about to what extent improvements were attributable to treatment or nonspecific effects.^{44,45}

Finally, the overall dropout rate in our study was 49%, with a 1-year dropout rate of 31%. The 1-year dropout rate was lower than the dropout rate found in a meta-analysis on dropout from psychological treatment for BPD (40%⁴⁶) but noticeably higher compared to the dropout rate observed in the other DBT-ST trial.¹⁴ No significant differences in dropout rates were observed between DBT and ST. The DBT 1-year dropout rate in our study was comparable to those reported in several meta-analyses (34% vs 23%-41%).⁴⁶⁻⁴⁹ However, the dropout rate for ST was markedly higher than in a previous study that investigated a comparable format of ST (1 year: 29% vs 18%, 2 years: 46% vs 26%).¹³ Dropout varied substantially across centers, with up to 75% of patients discontinuing ST in 2 centers. Our trial faced several challenges in the execution of the treatments in routine clinical practice at some participating centers, which may have contributed to the dropout rates, including high therapist turnover, organizational instability,⁵⁰⁻⁵² and ad hoc concerns raised by supervisors regarding the extent to which supervisory instructions and the treat-

ment protocol were implemented. As high dropout rates constitute a significant problem in BPD treatment, further research into factors that may predict dropout is warranted. One potential factor seems particularly understudied: instability in the team and organization.^{50,51}

Strengths and Limitations

This study has several strengths, including the comprehensive assessment of treatment response employing both self-report and observer-rated instruments with a 1-year follow-up period, a research group with balanced allegiance, the multicenter design, the comparable treatment dose across DBT and ST, and the high ecological validity ensured by routine clinical settings and minimal exclusion criteria. However, the study also has some limitations. First, considerable missing data on the reassessments may have biased the observed treatment effects, although the sensitivity analysis indicated that treatment effectiveness was not confounded by dropout. Second, only treatment adherence was assessed, not therapist competence, limiting insight into treatment quality.⁵³ Moreover, although treatment differentiation was observed in both individual and group sessions, inadequate interrater agreement warrants caution in interpreting findings for individual sessions. Third, this trial was conducted during the COVID-19 pandemic, which may have influenced the burden experienced by individuals with BPD.⁵⁴ Moreover, patients temporarily received treatment via videoconferencing, and treatment was interrupted during the first lockdown, although no confounding effects were found in the sensitivity analyses. Fourth, the DBT format studied in the present study differed from the standard format, with between-session consultation limited to the therapist's availability and a longer treatment duration that included a recently developed blended aftercare program.²⁶

Conclusion

This multicenter RCT found no significant differences in effectiveness between DBT and ST for outpatients with BPD,

with substantial improvements observed across outcomes in both treatments, consistent with findings from the previous DBT-ST trial.¹⁴ However, attributing these improvements to treatment and determining equivalence between

DBT and ST require further investigation. Additionally, future research may focus on understanding which treatment works best for whom and why, questions further explored as part of the BOOTS project.¹⁹

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